W-A-450/16
JANUARY 3, 1992

FEDERAL SPECIFICATION

DETECTOR, ACOUSTIC, PASSIVE / DISCRIMINATOR, SONIC

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1. SCOPE AND CLASSIFICATION.

1.1 Scope. The W-A-450C specification covers intrusion detection equipment which is deployed as a system to include, but is not to be limited to, detection of intrusions relating to protection of national security information, sensitive information and equipment, high dollar items, and weapons. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Unit and type. This specification covers Unit Number 1, Type XVI, Sonic Discriminator/Passive Acoustic Detector.

2. APPLICABLE DOCUMENTS.

The documents listed in W-A-450C/GEN, of the issues in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

3. REQUIREMENTS.

3.1 Qualification. The passive acoustic detector furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list. It shall conform to the minimum specifications required herein and shall have been tested and shall have passed the qualification tests specified in 4.1.

3.2 Temperature and humidity ranges.

3.2.1 Operating range. All equipment shall be capable of operating within the temperature and humidity ranges indicated below:

0 degrees C to +49 degrees C and relative humidity of 0% to 85%, non-condensing.

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All units shall withstand the tests in 4.1.3 through 4.1.5. Additional requirements may be imposed by the procuring entity.

3.2.2 Nonoperating range. All equipment which is not powered up shall be capable of withstanding, without damage, the temperature range indicated below:

-46 degrees C to 71 degrees C.

3.3 Dependability. All equipment supplied under this specification shall have a minimum Mean Time Between Failure (MTBF) of 10 000 hours.

3.4 Operation. The passive acoustic detector shall consist of receiver or microphone elements and necessary control circuitry to detect acoustic energy within the intrusion detection area. Usable and published detection patterns shall be included in the manufacturer's literature. Failure to supply this information shall be cause for non-compliance of the sensor. The detector shall, after initial setup, stabilize within five minutes after being powered on and shall also be designed so that nuisance alarms due to electromagnetic emissions from other equipment such as fluorescent lights or motors are prevented. Two types of passive acoustic detectors exist. The first is sensitive to acoustic energy at any frequency. The second is sensitive only to acoustic energy of certain frequencies. An audible test stimulus of a given frequency and intensity is transmitted to the passive acoustic detector. This test stimulus signal shall be in the frequency range necessary for detection by the detector being tested. If this test signal is of sufficient amplitude, the detector shall alarm. More detail on the test procedures is given in paragraph 4.1.1. Single-pole, double-throw alarm switch or relay and separate single-pole, double-throw tamper switch or relay connections shall be provided at a terminal strip for connection to external alarm reporting equipment. Alarm and tamper switches shall provide both normally open and normally closed contacts. Switch impedance shall be as follows:

Open state
- greater than 100 000 ohms.

Closed state
- For solid state relays - less than 50 ohms.
- For mechanical relays - 0.1 ohms or less.

3.4.1 Additional capabilities. The manufacturer is encouraged to provide information in their technical literature concerning performance data that would exceed the minimum requirements of this specification.

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3.5 Passive acoustic detector enclosure. The detector assembly shall be housed in an enclosure having a removable cover arranged to provide ready access for mounting, wiring, and alignment, and for sensitivity and other internal adjustments. Mounting holes shall be provided in the back of the enclosure in positions that are readily accessible (sufficiently displaced from all internal parts and the enclosure side) when all internal parts and wiring are in place. No external hardware or fasteners except the cover fasteners shall be used.

3.6 Cover. The cover shall be designed such that any attempt to access the interior shall cause a tamper condition. The tamper circuits shall be in their secure state whenever the cover is in place and shall go to their alarm state when the cover is moved sufficiently to reveal the inside or to introduce a defeating device. The tamper switch shall be constructed of corrosion resistant materials and operate smoothly to prevent jamming.

3.7 Terminals. Screw terminals shall be provided for interconnecting wiring. Access to the terminals shall require removal of the enclosure cover.

3.8 Switch and/or relay assembly. Alarm and tamper switch and/or relay contacts shall be protected against environmental and excessive current conditions which may be damaging. Sustained currents up to 1.33 times the rated current rating shall not cause damage to the switch or contacts.

3.9 Size, weight and power consumption. The size, weight and power consumption of each unit shall be the minimum practicable, consistent with meeting the requirements of this specification and shall be stated in the manufacturer's literature.

3.10 Maintainability. The detector shall be designed and constructed in such a manner as to minimize maintenance and repair time. To the maximum extent practicable, items shall be of a modular, unitized construction. Components shall be so arranged and assembled that they are readily visible and accessible to maintenance personnel. Specialized training is not required to maintain this equipment. Field maintenance shall consist of trouble shooting and isolation of failures to the device or replaceable sub-assembly, and replacement of the defective device or sub-assembly. Preventive maintenance may be required. Adjustments shall require no disassembly of the detector beyond mere removal of the cover.

3.11 Life cycle. The detector alarm and tamper switches and/or relay contacts shall be rated at not fewer than 1 000 000

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operations under a load of 250 mA driven from a 48 VDC source. Contact resistance shall not change more than +20% from the initial reading.

3.12 EMI compatibility. Units shall operate normally when subjected to the testing of 4.1.6. The testing shall neither inhibit valid alarms nor generate invalid alarms.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Tests.

4.1.1 Performance test. The unit being tested shall be mounted two meters away directly in front of a calibrated impulse sound source as shown in figure 1. This sound source shall be capable of reproducing acoustic energy from 100 to 25 000 cycles per second. If the unit is a sound discriminator such as a glass-break detector, only the frequencies associated with breaking glass shall be reproduced. The amplitude of this audio signal shall be of a magnitude to simulate glass breaking at the manufacturer's claimed maximum distance of detection. The detector should go into alarm under these conditions.

If the detector is not a sound discriminator, it must still alarm in response to the glass-break stimulus. In addition, the detector must alarm on stimulus at 125, 250, 500, 1000, and 2000 Hertz. These five additional stimuli shall be of the same magnitude at the detector as the glass-break stimulus. Both sound discriminator and non-sound discriminator detectors must be tested for area of coverage (field of view) if such a value is given. This is done by repeating the above tests with the sound source off axis to simulate a stimulus on the edge of the detection envelope as shown in figure 1. The measured detection pattern should be within 5% of the claimed coverage. All of these tests are to be made with the sensitivity controls at maximum.

4.1.2 Stability test. The unit submitted for qualification testing shall perform its intended function in a stable environment free of stimuli that would initiate a nuisance alarm. The test period shall be 90 days of continuous operation with any adjusted controls set to provide the specified security. The test period shall commence after the unit has been installed in accordance with the manufacturer's instructions and after a "shakedown" operational period of not more than seven consecutive days.

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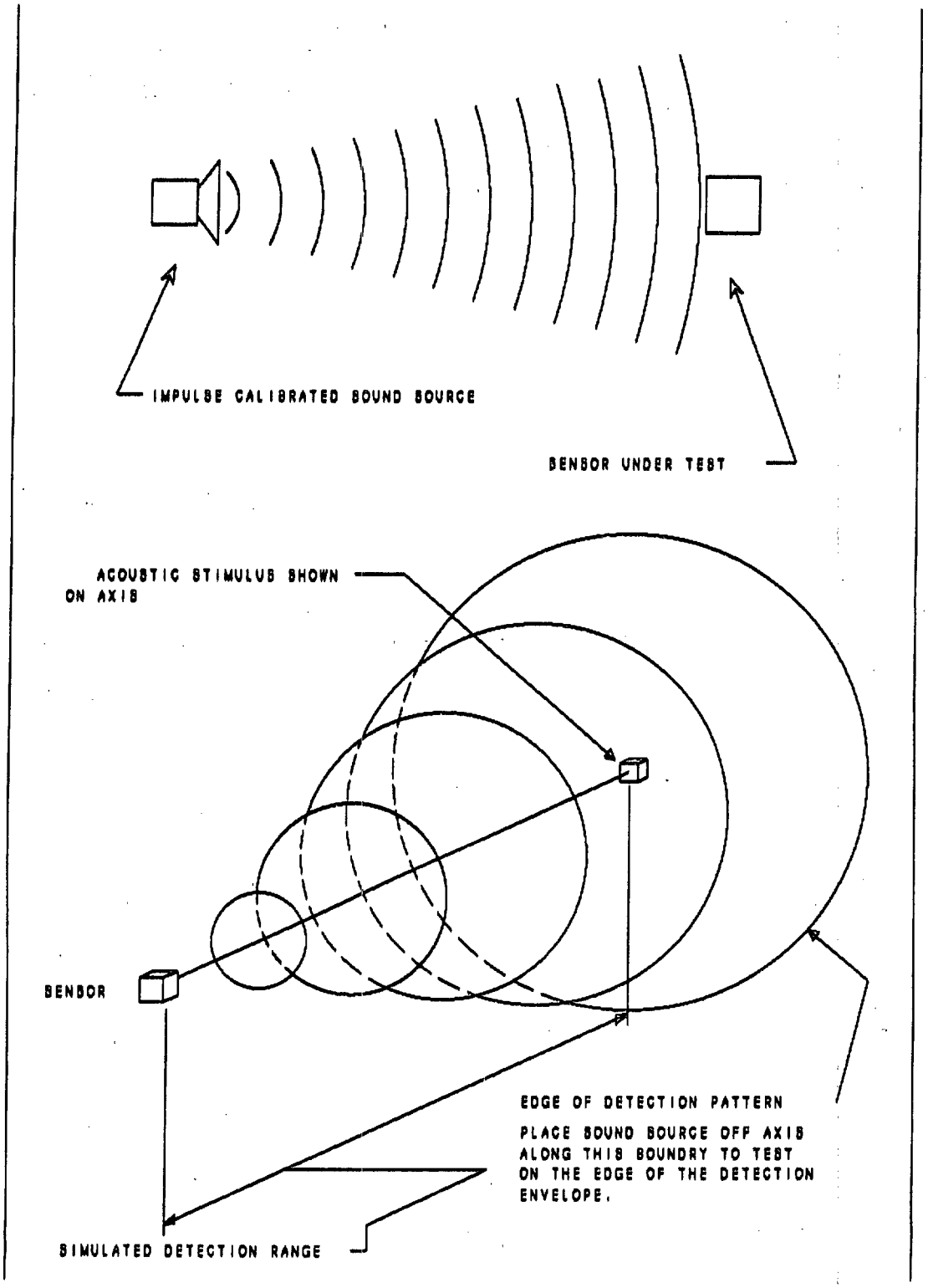


Figure 1

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4.1.3 High temperature test. Units submitted for qualification shall be energized for a period of four hours at 49 degrees C. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 49 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.4 Low temperature test. Units submitted for qualification shall be energized for a period of four hours at 0 degrees C. The unit shall then be tested as specified in 4.1.1 with the temperature maintained at 0 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.5 Humidity test. Units submitted for qualification shall be energized for a period of four hours with the humidity maintained at 85% at a maintained temperature of at least 26 degrees C. The test chamber shall be vented to the atmosphere to prevent the accumulation of moisture. The unit shall be tested as specified in 4.1.1 with the temperature maintained at 26 degrees C and the humidity maintained at 85%. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature and humidity must be stabilized for 5 minutes before the test period commences.

4.1.6 EMI compatibility test. Units shall be tested for compliance with 3.12 in accordance with the applicable tests of MIL-STD-462 with consideration of the following paragraphs:

4.1.6.1 Test chamber. All testing shall be performed in an EMI test chamber.

4.1.6.2 Conducted electromagnetic susceptibility test. The units shall be subjected to power line conducted interference in accordance with test methods CS01, CS02, and CS06 of MIL-STD-462. Maintain a test level of 1 or 3 Volts RMS as required over the entire frequency range.

4.1.6.3 Radiated electromagnetic susceptibility test. The units shall be subjected to radiation in accordance with test method RS03 of MIL-STD-462. Use an electric field of 1 Volt per meter for frequencies in the range of 14 Khz. to 2 Mhz. and 3 Volts per meter for frequencies in the range of 2 Mhz. to 12 Ghz.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking.

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Preservation, packaging, packing and marking shall be in accordance with ASTM D3951. The level of packing shall be A, B, or C, as specified (see 6.2).

5.1.1 Civil agency marking. In addition to any special markings (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Sensor systems constructed with components qualified under this specification are intended to supplement or support protective personnel (e.g., guards) for areas requiring special security measures. These properly assembled systems are resistive to neutralization or to compromise by covert or surreptitious attack. Guidance in selecting a particular item for a particular security situation is not intended and is beyond the scope of this document. Qualification approval merely means that the item meets the specification requirements at the time it was tested for qualification.

6.2 Ordering data. The purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type of coverage (detection pattern) required (where applicable).
(c) Quantities required.
(d) Spare parts required.
(e) Destination inspection required.
(f) Levels of packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).
(h) Shipping schedule required.

6.3 Qualification. Regarding products which require qualification under this specification, awards will be made only for products which have been tested and approved for inclusion on the applicable Federal Qualified Products List prior to the time set for opening of bids whether or not such products have actually been listed. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is:

General Services Administration

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Information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

6.4.1 Alarms.

6.4.1.1 Valid alarm. A valid alarm is any alarm generated by stimuli which the detector is designed to recognize and which satisfies the detector's stated alarm criteria.

6.4.1.1 Invalid alarm. Any other alarm.

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